

Week 2, Lecture 1: Working with Functions

Instructor notes – Part 1: prep reading, Part 2: podium cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the podium copy.

Source material: Bolker Ch. 3, sections 3.2-3.4 (“Deterministic functions for ecological modeling”) almost verbatim in places – this lecture leans harder on the assigned reading than Week 1’s did, because today’s content is a hard prerequisite for a specific Lab 2 question (see below). Also mined: this morning’s WOD (`wod/WOD_Week2_Lecture1_RSyntaxAndNotation.Rmd`), Lab 1 Act 2 (Ricker derivative), and Lab 2 Acts 1-2 (grid search, flexible Ricker, Holling IV).

Where this sits in the week, and a hard dependency

- Monday (this lecture): properties of deterministic functions – limits, derivatives, critical points – as tools, not yet the bestiary itself (that’s Wednesday).
- Monday AM (WOD, already done): bug-hunts in R function syntax (argument order/matching) plus deriving and coding a seasonal sine function – both are direct rehearsals for today.
- Tuesday (Lab 2): Act 1 is a grid-search optimizer for the Ricker function (algorithms, not functions per se – a callback to Week 1). **Act 2 explicitly instructs students to use “the same ‘throw out the lower-order terms’ trick from Monday’s lecture”** to find the limiting behavior of the Holling type IV function as $x \rightarrow \infty$. That phrase is not a coincidence – it’s this lecture. If Section 2 below (limits) doesn’t land clearly today, that Lab 2 question will stall.
- Wednesday: Bestiary of functions (Bolker 3.5) – the specific named catalog (power, exponential, hyperbolic, sigmoid...). Today builds the toolkit; Wednesday populates it.

Big picture (bridge from Week 1)

One-sentence bridge: last lecture was about *how* we get from a candidate model to an answer (algorithms vs. closed form); today is about *reading* a candidate function itself – what it does at the extremes, where it peaks, what its parameters mean – before you ever fit it to data.

Recall from Lab 1 Act 1: students already have an informal shape vocabulary (monotonic increasing/decreasing, saturating, hump-shaped, roughly flat) from eyeballing the tadpole

data. Today's tools let you *prove* a shape claim algebraically instead of just eyeballing a curve – the same distinction as Bolker 3.2/3.3 vs. 3.4: numerically (plug in values, plot it, `curve()`) gives you a picture; analytically (algebra and calculus) tells you *why* the picture looks that way, and lets you reason about regions you didn't plot.

Limits: throw out the lower-order terms (the required section)

This is Bolker's own phrasing (3.4.1), worth keeping close to verbatim since Lab 2 Act 2 quotes it back: *"The basic principle is to throw out lower-order terms. As x grows, it will eventually grow much larger than the largest constant term in the equation. Terms with larger powers of x will dwarf smaller powers, and exponentials will dwarf any power. If x is very small then you apply the same logic in reverse; constants are bigger than (positive) powers of x , and negative powers... are bigger than any constants."*

In short, a hierarchy of "who wins" as $x \rightarrow \infty$:

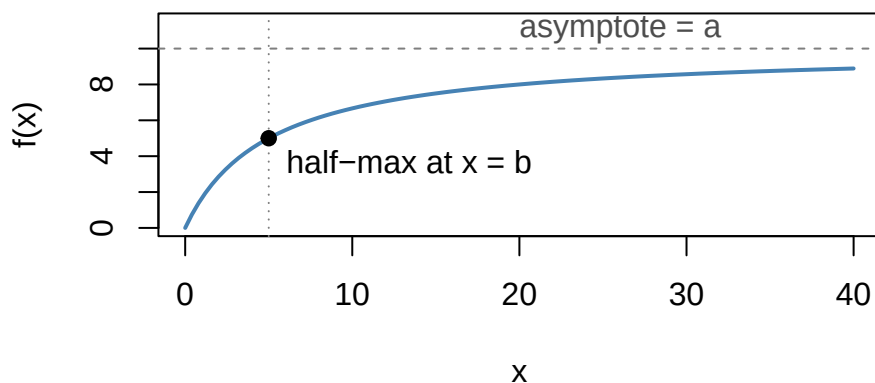
$$\text{exponential} \gg x^{\text{large power}} \gg x^{\text{small power}} \gg \text{constant}$$

and the *same* hierarchy reverses as $x \rightarrow 0$: constants beat positive powers of x , and negative powers of x (i.e. $1/x$, $1/x^2$, ...) beat everything.

Worked example (Bolker's own – Michaelis-Menten), do this on the board:

$$f(x) = \frac{ax}{b+x}$$

- As $x \rightarrow \infty$: $x \gg b$, so $b+x \approx x$, so $f(x) \approx ax/x = a$ – the curve levels off at the constant a (a horizontal asymptote).
- As $x \rightarrow 0$: $b \gg x$, so $b+x \approx b$, so $f(x) \approx ax/b$ – a straight line through the origin with slope a/b .



Sticking point to flag: students will want to “plug in infinity.” Head that off explicitly – this is qualitative dominance reasoning (which term is negligible), not literal substitution. Also flag that the *same* parameter can play opposite roles at the two ends: b is negligible as $x \rightarrow \infty$ but dominant as $x \rightarrow 0$.

Fallback tool (mention, don’t dwell): for fraction limits where both numerator and denominator vanish or blow up together (an indeterminate $0/0$ or ∞/∞ form, so the dominance trick doesn’t immediately apply), L’Hopital’s rule says $\lim \frac{a(x)}{b(x)} = \lim \frac{a'(x)}{b'(x)}$ – take the derivative of the top and bottom separately, not the derivative of the whole fraction. One line, not a worked example today.

Do not solve Holling IV’s limit here. $y = ax^2/(b + cx + x^2)$ is literally Lab 2 Act 2’s question 1c – comparing the highest powers of x in numerator (x^2) and denominator (x^2) is the same dominance trick, one level harder (both leading terms are degree 2, so the constant a from the ratio of leading coefficients survives). Let students discover that tomorrow; today’s example (Michaelis-Menten) is deliberately a different, simpler rational function so the technique transfers without the answer being pre-given.

Derivatives and critical points (recap + formalize)

Students already computed one derivative and one critical point by hand in Lab 1 Act 2 – this section names what they did, generalizes it, and contrasts it with a monotonic function that has no peak at all.

Recap, don’t re-derive from scratch: the Ricker function $y = axe^{-bx}$ (Lab 1’s running example: $a = 0.9$, $b = 1/12$). Product rule:

$$\frac{dy}{dx} = ae^{-bx} + ax(-be^{-bx}) = a(1 - bx)e^{-bx}$$

Setting this to zero: since $ae^{-bx} \neq 0$, the only way the product is zero is $1 - bx = 0$, i.e. $x^* = 1/b$.

```
a_r <- 0.9; b_r <- 1/12
x_star <- 1 / b_r
peak_height <- a_r * x_star * exp(-b_r * x_star)
x_star
```

```
## [1] 12
```

```
peak_height
```

```
## [1] 3.973098
```

That’s $x^* = 12$ mm and peak height 3.97 – matching Lab 1 Act 3’s table exactly (TBL = 12, Ricker predicted = 3.97). This is the payoff of the “critical point” concept: it’s not new math, it’s a name for exactly the calculation they already did.

Formalize: a critical point is any x where $f'(x) = 0$ (or is undefined) – a candidate for a peak, a trough, or (at a saddle point) a flat spot that isn't either. To classify which, plug the critical x -value back into f itself (peak/trough height) and reason about the sign of f' on either side (or use the second derivative).

Contrast with a monotonic function: try the same exercise on Michaelis-Menten, $f(x) = ax/(b + x)$. Its derivative (quotient rule) is $f'(x) = ab/(b + x)^2$ – always positive (a sum of squares in the denominator can't be negative), so the function is *monotonic increasing* everywhere and *never* has a critical point. That's exactly why it saturates toward an asymptote instead of peaking and coming back down – contrast this shape against Ricker's hump directly, on the board, side by side.

Sticking point to flag: “critical point” (where $f'(x) = 0$) is easy to confuse with “root” (where $f(x) = 0$). The Ricker function has both: it's zero at $x = 0$ (a root) and peaks at $x = 1/b$ (a critical point) – two different questions with two different answers from the same function.

R convention: x first, then parameters (with defaults)

Directly from Bolker's R supplement (3.3.1, p. 104): “When you write such an R function, you would typically make the input value (x) be the first argument, followed by all of the other parameters. It often saves time if you assign default values to the other parameters.”

This is exactly why `ricker <- function(x, a, b)` is the convention used everywhere in the labs (and why WOD 1a's `function(a, b, x)` was buggy this morning – it broke that convention on purpose). Point this out explicitly: the WOD wasn't testing an arbitrary rule, it was testing *this* rule, which is about to matter again.

Why it matters going forward, concretely: `sapply(xvec, ricker)` only works cleanly if x is the first positional argument – `sapply` always feeds the vector element into the first open slot. Also preview: Lab 2 Act 2 asks students to write `flex_ricker` and `holling4` “following the same pattern as the `ricker` function (x first, then parameters)” – today is where that instruction gets its justification, not just its statement.

Scale parameters, briefly (Bolker 3.4.2): when a parameter appears as bx or x/c , its job is to stretch or shrink the curve along the x -axis without changing its shape – Bolker calls this a *characteristic scale*. b has units of inverse- x ; $c = 1/b$ has the same units as x and is often easier to interpret (e.g. “the peak is at $x = 12$ ” instead of “ $b = 1/12$ ”). This is exactly what Lab 2 Act 2's flexible Ricker does: it reparameterizes b as p (the peak location directly) precisely so the parameter is easier to interpret and to set. One sentence today is enough – Act 2 does the work of actually reparameterizing.

Take-home message

- Two ways to know a function: numerically (plot it, plug in numbers) and analytically (limits, derivatives) – today was almost entirely the analytical toolkit.

- Limits at the extremes: throw out lower-order terms. Same hierarchy, reversed at the two ends (exponential $\gg$ big power $\gg$ small power $\gg$ constant $\gg$ negative power, as $x \rightarrow \infty$; flip it for $x \rightarrow 0$).
- Critical points ($f'(x) = 0$) find peaks/troughs; monotonic functions (no sign change in f') never have any.
- R convention (**x** first, params after, with defaults) isn't arbitrary style – it's what makes `sapply()` and vectorized code work cleanly, and it's the same convention Lab 2 asks you to follow when writing new bestiary members from scratch.

Roadmap tie-in

- Wednesday: the bestiary itself (Bolker 3.5) – linear, power, exponential, hyperbolic/Michaelis-Menten, sigmoid families, each characterized using exactly today's tools (limits, critical points, scale parameters).
- Tuesday's Lab 2 Act 2 needs today's limit technique directly (Holling IV, question 1c) – flagged above, worth a one-line reminder at the end of lecture.
- Later in the semester: “throw out lower-order terms” reasoning resurfaces informally whenever we ask what a fitted model predicts far outside the range of the data (extrapolation risk) – worth a passing mention now, not a digression.

Discussion seed (if time remains)

Ask: *pick any function from your own field that saturates (levels off) as some predictor gets large. What's the mechanism that caps it – what in the real world plays the role of “a” (the asymptote) in Michaelis-Menten?* Good prompt for surfacing intuition about *why* saturation happens biologically, not just how to compute it. (If it comes up: “b” is a different question – not what caps the curve, but how fast it gets there.)

PART 2 – PODIUM CHEAT SHEET

Podium copy – terse on purpose. Full derivations, rationale, and code: see Part 1 above.

0. Open (~2 min)

- Last time: *how* we get an answer (algorithm vs. closed form)
- Today: *reading* a function itself – extremes, peaks, parameters
- Recall Lab 1’s shape vocabulary (monotonic / saturating / hump) – today we prove those claims, not just eyeball them

1. Two ways to know a function (~3 min)

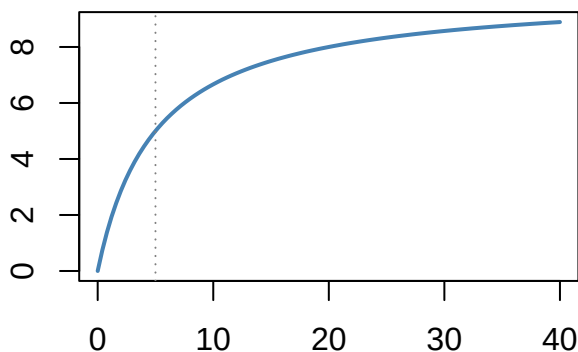
- **Numerically:** plug in values, `curve()`, plot it – a picture
- **Analytically:** limits + derivatives – *why* the picture looks that way
- Today = almost entirely analytical

2. Limits: throw out the lower-order terms (~12-15 min)

The rule (Bolker's own words – write it up):

- as $x \rightarrow \infty$: exponential $\gg$ big power $\gg$ small power $\gg$ constant
- as $x \rightarrow 0$: reverse it – constant $\gg$ positive power $\gg$ nothing beats a negative power ($1/x$, $1/x^2$, ...)

Board example – Michaelis-Menten: $f(x) = \frac{ax}{b+x}$



- $x \rightarrow \infty$: $b+x \approx x \Rightarrow f(x) \approx ax/x = a$ (asymptote)
- $x \rightarrow 0$: $b+x \approx b \Rightarrow f(x) \approx ax/b$ (line, slope a/b)
- **Not** “plug in infinity” – dominance reasoning, term by term
- b is negligible at one end, dominant at the other – same parameter, opposite roles
- Fallback only, don't dwell: L'Hopital's rule for $0/0$ or ∞/∞ forms – derivative of top over derivative of bottom
- **Do not solve Holling IV today** – that's Lab 2 Act 2, question 1c, tomorrow. Michaelis-Menten teaches the same trick without giving away the answer.

3. Derivatives and critical points (~8-10 min)

Recap (already done in Lab 1 Act 2), don't re-derive: Ricker $y = axe^{-bx}$, product rule $\rightarrow dy/dx = a(1 - bx)e^{-bx}$

- critical point: $1 - bx = 0 \Rightarrow x^* = 1/b$
- with $a = 0.9$, $b = 1/12$: $x^* = 12$, height = 3.97 – matches Lab 1 Act 3's table exactly
- **critical point:** name for exactly what they already computed

Contrast – Michaelis-Menten's derivative (quotient rule): $f'(x) = ab/(b+x)^2$ – always positive $\rightarrow$ monotonic, **no** critical point, saturates instead of peaking

- **Sticking point:** critical point ($f' = 0$) $\neq$ root ($f = 0$) – Ricker has both, at different x (root at $x = 0$, peak at $x = 1/b$)

4. R convention + scale parameters (~5 min)

- Bolker's own rule: **x** first argument, params after, with defaults
- *this is why* `function(a, b, x)` in this morning's WOD 1a was buggy
- matters for `supply()`: vector element fills the *first* open slot
- **preview:** Lab 2 Act 2 asks you to write new bestiary functions “following the same pattern (**x** first, then parameters)”
- **scale parameter:** bx or x/c stretches/shrinks the x -axis; Lab 2's flexible Ricker reparameterizes $b \rightarrow p$ (peak location directly) for exactly this reason

5. Roadmap (~2 min)

- Wednesday: the bestiary itself (linear, power, exponential, hyperbolic, sigmoid) – same tools, now cataloging specific families
- **Reminder before they leave:** Lab 2 Act 2 needs today's limit trick for Holling IV

If time: discussion seed

- *Pick a function from your own field that saturates. What plays the role of “a” (the asymptote) – what caps it, mechanistically?*